

Tilli Measures!

The 360-degree Foundational Skills Dataset

Unlocking longitudinal, multi-informant insights into cognitive and social-emotional development for non-WEIRD populations.

[Explore The Data Dictionary](#)
[Access Data Repository](#)

The Problem: "Data Blindness" in AI

1. Sparse Outcomes

Most datasets focus only on static final scores, ignoring the "process" of how a child learns.



2. Isolated Data

Research is often siloed either looking at classroom performance or home environments, but never triangulating both.



3. Expensive Protocols

Standardized assessments require high-cost, in-person trained assessors, leaving resource-poor communities out of the AI evolution.



Current educational AI models are predominantly trained on data from

W.E.I.R.D.

Western, Educated, Industrialized, Rich, and Democratic populations.

This creates a

"data blind spot" for children in

South Asia, the Middle East, and other diverse contexts.

We provide a unified view of child development!

Across 12 core competencies, categorized into two critical domains

6 Cognitive Skills



Attention
The ability to focus on relevant information and sustain concentration on a task.



Working Memory
The capacity to hold and manipulate information over short periods.



Cognitive Flexibility
The ability to shift between tasks, perspectives, or strategies as needed.



Inhibition Control
The ability to regulate impulses and resist distractions.



Response Suppression
The capacity to pause and think before acting.



Planning and Organization
The ability to set goals, plan steps, and structure tasks effectively.



Emotion Awareness
The ability to recognize and understand one's own emotions.



Emotion Regulation
The capacity to manage and respond to emotions appropriately.



Relationship Skills
The capacity to build and maintain positive interactions with others.



Metacognition
The ability to reflect on and understand one's own thinking processes.



Empathy
The ability to understand and share the feelings of others.



Critical Thinking
The ability to analyze and evaluate information to form reasoned judgments.

6 Social-Emotional Learning (SEL) Skills

OTHER DOMAIN AND PARAMETERS

Academic Domain

Child-Level Predictors

Ecological Predictors

Academic Domain

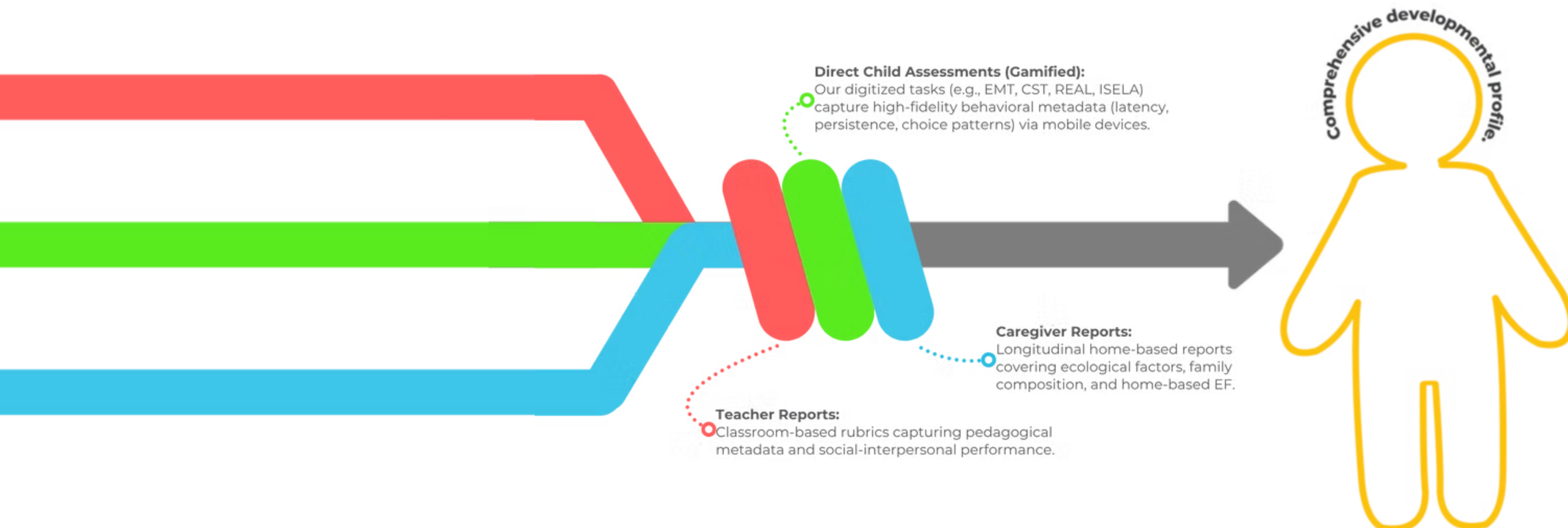
- Math Scores
- Literacy Scores

Captures core learning outcomes through math and literacy. Math reflects problem-solving and numerical understanding, while literacy measures reading, comprehension, and expression—together indicating overall academic proficiency.



How We Measure: The Triadic Assessment Protocol

We move beyond single-informant bias by capturing three perspectives for every child, using a digitized, voice-supported, and localized protocol:



Note: All data is de-identified, privacy-first, and delivered via low-bandwidth messaging platforms to ensure inclusivity.

Demographics & Ecology

Equity markers, home environment, family structure, and special needs status.

Direct Assessments

Behavioral interaction streams, raw latency, task-switching performance, and foundational literacy/numeracy/readiness.

Caregiver Report

Home-based emotional symptoms, behavioral outcomes, and executive function ratings.

Teacher / Classroom Report

Pedagogical context, concentration/distractibility ratings, and interpersonal assessments.

What We Measure: Data Structure

Our dataset is composed of four interoperable CSV files linked by a unique subject_id.

Our high-density behavioral streams enable researchers to build the next generation of AI!

AI Use Cases:

From Research to Action



1/ Predictive Intervention

Identify Grade 3 cognitive signatures that serve as early warning signs for academic struggle.

2/ Context-Aware Profiling

Use AI to weigh conflicting reports from parents vs. teachers, turning "discrepancies" into signals for better child support.

3/ Synthetic Learner Profiles

Train Generative AI (LLMs/LRMs) to simulate diverse, realistic student profiles for teacher training and curriculum stress-testing.

4/ Algorithmic Fairness Audits

Use our localized dataset to audit and improve the regional robustness of global SEL models.

Resources and More!

Technical Documentation

This guide is intended for software developers who are familiar with how web applications are built.

[Link to PDF Manuals](#)

GitHub

This guide is intended for developers using GitHub who are familiar with how web applications are built.

[Link to Open-source Toolkits](#)

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About the Team

We are a Research-Practice Partnership (RPP) involving UNICEF, Tilli Kids, and global educators committed to open-source learning engineering.

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Research
Shop
Privacy Policy
FAQs

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Our Team
Events
Blogs

Our Socials

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Instagram
X (formerly Twitter)
YouTube
Facebook

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Hello! Kavindya here from Tilli. Let me know how I can help